10.1 Lines and Segments That Intersect Circles

Name _____ Period ____ Date _____

Geometry 10.1 (30-minute plan) · Student sheet

I can…

- name the special segments and lines of a circle.
- use the fact that a tangent is perpendicular to the radius at the point of tangency.
- use the fact that two tangent segments from the same point are congruent.

Vocabulary · fill in the missing word

Chord — a segment whose _____ are on a circle

Secant — a line that _____ a circle in two points

Tangent — a line in the plane of a circle that _____ the circle in exactly one point

Point of tangency — the point where a tangent line _____ the circle

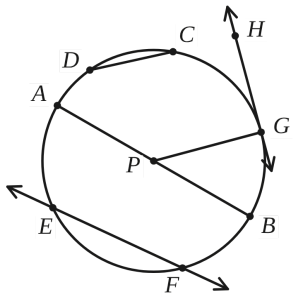
Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

In circle P, name each of the following.

a. a chord b. a diameter c. a radius

d. a secant e. a tangent f. a point of tangency

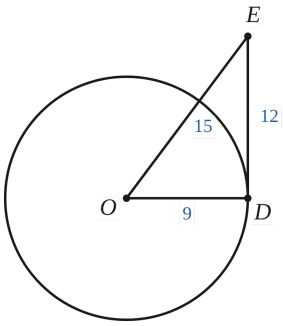


Tangent Line to Circle Theorem: a line is tangent to a circle if and only if it is _____ to a radius at its endpoint on the circle.

Testing for a tangent: use the Converse of the _____ Theorem. If $r^2 + (\text{segment})^2 = d^2$, the line is tangent.

Example 2

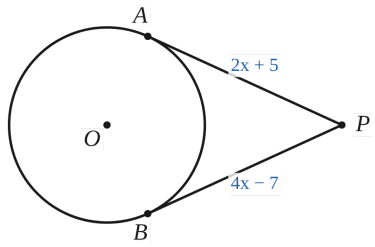
Is $\overline{ED}$ tangent to circle O? Explain.



External Tangent Congruence Theorem: tangent segments from a common external point are _____.

Example 3

$\overline{PA}$ and $\overline{PB}$ are tangent to circle O. Find the value of x. Then find PA.



Part 2 · Partner practice

Partner practice

Two tangent segments from the same point have lengths 3x + 2 and 5x − 8. Find x and the length of each segment.